

Your Code

```
1 def generate_permutations(s, restrictions):
2     n = len(s)
3     used = [False] * n
4     current = []
5     result = []
6     forbidden = set()
7     for a, b in restrictions:
8         forbidden.add((a, b))
9         forbidden.add((b, a))
10    def backtrack():
11        if len(current) == n:
12            result.append("".join(current))
13            return
14        for i in range(n):
15            if used[i]:
16                continue
17            if current:
18                if (current[-1], s[i]) in forbidden:
19                    continue
20            used[i] = True
21            current.append(s[i])
22            backtrack()
23            current.pop()
24            used[i] = False
25
26    backtrack()
```

Input

stdin input

Output

```
Valid permutations:
ACBD
ACDB
ADBC
ADCB
BCAD
```

```
23         current.pop()
24         used[i] = False
25
26     backtrack()
27     return result
28 s = "ABCD"
29 restrictions = [('A', 'B')]
30
31 valid = generate_permutations(s, restrictions)
32
33 print("Valid permutations:")
34 for p in valid:
35     print(p)
36
37 print("Total valid permutations:", len(valid))
```

✔ Submitted